Two-point Feedback for Learning One-Sided Lipschitz Functions

Shiliang Zuo

Abstract

We study adversarial online learning on $[0,1]$ when every loss is one-sided Lipschitz and the benchmark is the best fixed action in hindsight. The learner may select two actions simultaneously, observes both losses, and pays their arithmetic mean; thus the second observation is not a free probe. We prove an upper bound

$$O\Big(\sqrt{T}\,(1+\log T)^{5/4}\Big)$$

for this two-paid-observation model and a lower bound $\Omega(\sqrt{T}\log T)$, leaving only a factor $(\log T)^{1/4}$. The upper bound uses a dyadic tree, a square-root regularizer on subtree masses, a small log barrier that controls finite updates, and a nonuniform tree-energy inequality. The lower bound hides a dyadic path in a threshold family: the best fixed action collects signal at every scale, while any simultaneous pair reveals only a three-bin statistic with $O(1/T)$ average information per round. Under full information we give a dyadic asymmetric learner with optimal $\Theta(\sqrt{T})$ regret. Finally, we show that the two-paid-point protocol has a direct mechanism-design interpretation: for posted pricing, every queried pair is implemented exactly by an incentive-compatible cumulative lottery menu with two positive thresholds.

1 Introduction

Many online decisions are ordered. A seller selects a price, a classifier chooses a threshold, or a controller chooses an operating point. In such problems the loss can be highly discontinuous, yet moving in one direction cannot improve the loss too quickly. We study the adversarial online-learning class

$$\mathcal{G}_{\mathrm{OSL}} = \{\ell : [0,1] \to [0,1] : \ell(x) - \ell(y) \le y - x \text{ for all } x \le y\},$$

with regret measured against the best fixed action in hindsight.

The central question is how much is gained by one additional observation when that observation must also be paid for. On every round our two-point learner chooses two actions simultaneously, observes both losses, and incurs their arithmetic mean. This differs from decoupled exploration models in which an extra query is free. The distinction is important in economic applications: querying a second price must itself correspond to an implementable transaction.

The feedback hierarchy is sharp at its two endpoints and nearly sharp in the middle:

Feedback	Regret against best fixed action	Status
one paid point	$\Theta(T^{2/3})$	known benchmark
two simultaneous paid points	$\Omega(\sqrt{T}\log T)$ to $O(\sqrt{T}(\log T)^{5/4})$	this paper
full information	$\Theta(\sqrt{T})$	this paper

Thus a second paid observation changes the horizon exponent from $2/3$ to $1/2$, but a logarithmic multiscale cost remains; full information removes that cost.

Two-point upper bound. Our main algorithmic result is

$$\mathbb{E}\mathrm{Reg}_T = O\Big(\sqrt{T}\,(1+\log T)^{5/4}\Big).$$

The learner discretizes the interval on a dyadic grid and maintains a distribution over leaves. The regularizer has two parts: a square-root potential summed over all subtree masses, which provides the correct multiscale geometry, and a small log barrier on the same masses, which keeps finite importance-weighted updates feasible even when the distribution becomes highly nonuniform. A random round either duplicates one leaf or compares the two children of one internal node. The paid marginal is corrected by a finite-chord term so that the potential increase and the exploration cost cancel in expectation. The remaining stability term is controlled by a new nonuniform energy inequality of order $L^{3/2}$ for a depth-L tree. Balancing this with the comparator cost yields the exponent $5/4$ on the logarithm.

Two-point lower bound. We prove that every simultaneous two-point learner suffers

$$\mathbb{E}\mathrm{Reg}_T = \Omega(\sqrt{T}\log T).$$

The hard instance already lies in a threshold family. A hidden dyadic leaf determines a nested path of Haar perturbations. At the hidden comparator, signals from all $L = \Theta(\log T)$ scales add coherently. A fixed queried interval, however, sees only a bounded average square increment of the integrated Haar process. Two threshold observations reveal exactly a three-bin statistic, and its average KL divergence from the null is only $O(1/T)$ per round. Hence the learner cannot identify the multiscale path quickly enough to capture the comparator's cumulative $\Theta(\sqrt{T}\log T)$ advantage.

Full information. With full feedback, every dyadic comparison can be updated on every round. The one-sided Lipschitz condition gives an asymmetric local range: at a node of width w, moving to the right can improve reward by at most w but can hurt by order one. We give a two-action primitive with regret $O(\sqrt{wT})$. Summing along one root-to-leaf path gives a geometric series and hence $O(\sqrt{T})$ regret. A two-linear-loss construction gives the matching lower bound.

Mechanism-design interpretation. Posted pricing is a canonical member of the class. For buyer value v, the revenue $p\mathbf{1}\{p \le v\}$ induces an OSL loss. A generic second oracle query would not by itself be a valid economic action. We show that any simultaneous pair of queried prices can instead be offered as an incentive-compatible cumulative lottery menu with two positive thresholds. The chosen menu option reveals both threshold bits, while the seller's payment is exactly the average revenue of the two queried posted prices. The abstract paid-feedback guarantee therefore yields a native mechanism with the same regret.

Relation to prior work. The one-point $\Theta(T^{2/3})$ benchmark contains adversarial posted pricing as a subclass; the classical posted-price problem of Kleinberg and Leighton (2003) gives the matching lower-bound scale, while uniform discretization plus a finite adversarial bandit algorithm gives the upper bound. One-sided Lipschitz structure has also been studied in a different repeated-optimization regime in which the same unknown objective is queried across rounds; Duetting et al. (2023) obtain logarithmic-in-logarithm regret in that stationary, noiseless setting. Our setting is adversarial across time, and the learner's observations are tied directly to the loss it pays. This paid-feedback constraint is the source of the multiscale reconstruction problem studied here.

Organization. Section 2 defines the model. Section 3 gives the two-point algorithm and upper bound, Section 4 gives the multiscale lower bound, and Section 5 treats full information. Section 6 gives the pricing implementation. All substantial proofs are deferred to the appendices.

2 Model and preliminaries

An oblivious adversary fixes losses $\ell_1, \dots, \ell_T \in \mathcal{G}_{\mathrm{OSL}}$, where

$$0 \le \ell_t(x) \le 1, \qquad \ell_t(x) - \ell_t(y) \le y - x \quad (x \le y). \tag{1}$$

The learner competes with the best fixed point in $[0,1]$.

One point. On round t, the learner chooses $X_t \in [0,1]$, incurs $\ell_t(X_t)$, and observes only this value. The expected regret is

$$\mathbb{E}\sum_{t=1}^{T} \ell_t(X_t) - \min_{x\in[0,1]} \sum_{t=1}^{T} \ell_t(x).$$

The minimax rate is $\Theta(T^{2/3})$: discretizing to K points gives $O(\sqrt{KT} + T/K)$ via a standard adversarial bandit algorithm (Auer et al., 2002), optimized at $K \asymp T^{1/3}$; adversarial posted pricing supplies a matching lower bound (Kleinberg and Leighton, 2003).

Two simultaneous paid points. Before seeing any current-round feedback, the learner chooses a pair $(B_t, C_t) \in [0,1]^2$, observes $\ell_t(B_t)$ and $\ell_t(C_t)$, and incurs

$$\frac{\ell_t(B_t) + \ell_t(C_t)}{2}. \tag{2}$$

Its regret is

$$\mathrm{Reg}_T^{(2)} := \mathbb{E}\sum_{t=1}^{T} \frac{\ell_t(B_t) + \ell_t(C_t)}{2} - \min_{x\in[0,1]} \sum_{t=1}^{T} \ell_t(x). \tag{3}$$

Both endpoints are paid; there are no free probes. The pair may contain a duplicate, $B_t = C_t$.

Full information. The learner chooses X_t before observing ℓ_t, pays $\ell_t(X_t)$, and then observes the entire current loss function. The resulting minimax regret is denoted $R_T^{\mathrm{full}}(\mathcal{G}_{\mathrm{OSL}})$.

Downward rounding. The OSL inequality gives the discretization direction that will be used repeatedly. If $q \le x$ and $x - q \le h$, then

$$\ell_t(q) \le \ell_t(x) + h. \tag{4}$$

Thus rounding a fixed comparator downward to a grid of spacing $1/K$ costs at most T/K.

Reward form. When convenient, write $r_t = 1 - \ell_t$. Then (1) becomes

$$r_t(y) - r_t(x) \le y - x, \qquad x \le y. \tag{5}$$

The asymmetric direction of this inequality is crucial in both the upper and full-information analyses.

3 Two paid observations: upper bound

We first state the main algorithmic result.

Theorem 3.1 (Two-point OSL upper bound)**.** *There is a simultaneous two-query paid-average learner such that, for every oblivious sequence $\ell_1, \dots, \ell_T \in \mathcal{G}_{\mathrm{OSL}}$,*

$$\mathbb{E}\mathrm{Reg}_T^{(2)} \le C\sqrt{T}\,(1+\log T)^{5/4} \tag{6}$$

for a universal constant C.

The construction is easiest to describe on a dyadic grid. Let $K = 2^L$ and place the actions

$$\left\{0, \frac{1}{K}, \dots, \frac{K-1}{K}\right\}$$

at the leaves of a complete binary tree. A node v at depth d has spatial width $w_v = 2^{-d}$, and p_v denotes the total current leaf probability in its subtree.

The regularizer is

$$\Psi_0(p) = -\sum_v \sqrt{w_v p_v}, \qquad \Psi(p) = \Psi_0(p) - \mu \sum_v \log p_v, \tag{7}$$

where the root barrier term is omitted and μ is a small multiple of the learning rate. Let

$$\Phi(s) = \max_{p \in \Delta_K^\circ} \{\langle s, p\rangle - \Psi(p)\}, \qquad p = \nabla\Phi(s). \tag{8}$$

The square-root part supplies multiscale curvature; the log barrier is used only to control finite updates and maintain a legal paid sampler at arbitrarily nonuniform states.

For an internal node v, let vL, vR be its children and define

$$\tau_v = -\frac{p_{vR}}{p_v}\mathbf{1}_{vL} + \frac{p_{vL}}{p_v}\mathbf{1}_{vR}. \tag{9}$$

At depth d, write

$$Z = \sum_{h=1}^{L} h^{-1/2}, \qquad \beta_d = \frac{1}{Z\sqrt{L-d}}, \qquad q_v = \beta_d \min\{p_{vL}, p_{vR}\}. \tag{10}$$

With probability q_v, the learner draws one endpoint from each child according to the current conditional distribution and queries the pair; the remaining probability is assigned to duplicate queries. A finite-chord correction changes the actual paid marginal from p to a nearby distribution π and simultaneously cancels the first-order cost of the importance-weighted update. The full formulas appear in Section A.

The proof has three ingredients.

Finite-chord cancellation. If a split at v observes the loss difference D, the score update is a multiple of $D\tau_v$. The OSL condition implies $D \in [-w_v, 1]$. Interpolating the dual potential over exactly this asymmetric chord yields a one-step inequality in which the cost of replacing p by the paid marginal π cancels. After telescoping, the regret is bounded by a comparator term plus a sum of curvature costs.

Barrier-controlled feasibility. The log barrier gives enough descendant curvature to control the entire finite chord, not merely an infinitesimal Hessian approximation. With

$$\mu = 32\eta L,$$

the duplicate probabilities remain nonnegative at every positive state. In fact each leaf i retains duplicate mass at least $p_i/4$. Thus the algorithm is a legal simultaneous paid sampler even when some subtree masses are arbitrarily small.

Nonuniform tree energy. Let G_0 be the inverse-Hessian quadratic form of Ψ_0 on the simplex tangent space. The key static estimate is

$$\mathcal{A}_0(p) := \sum_{v \text{ internal}} \frac{w_v \, \tau_v^\top G_0 \tau_v}{\beta_{\mathrm{depth}(v)} \min\{p_{vL}, p_{vR}\}} \le 128 L^{3/2} \tag{11}$$

for every strictly positive leaf distribution p. This is the step that improves the previous logarithmic exponent: it remains valid far from the uniform state and requires no probability floor.

Combining the three ingredients gives, for a K-point grid and an interior smoothing parameter ε,

$$\mathbb{E}\mathrm{Reg}_T^{\mathrm{grid}} \le \frac{L+1}{\eta} + 64KL\log(1/\varepsilon) + 128\eta T L^{3/2} + \varepsilon T. \tag{12}$$

Rounding a continuous comparator adds T/K. Taking $\varepsilon = T^{-1}$, K to be the smallest power of two at least $\max\{2, \sqrt{T}/(1+\log T)\}$, and

$$\eta = \frac{1}{\sqrt{T}\,(L+1)^{1/4}}$$

yields (6). The complete construction and proof are in Section A.

4 Two paid observations: lower bound

The logarithmic penalty in Theorem 3.1 cannot be removed altogether.

Theorem 4.1 (Two-point OSL lower bound)**.** *There are universal constants $c > 0$ and T_0 such that for every $T \ge T_0$ and every randomized simultaneous two-point paid-average learner, there is a deterministic oblivious sequence $\ell_1, \dots, \ell_T \in \mathcal{G}_{\mathrm{OSL}}$ satisfying*

$$\mathbb{E}\mathrm{Reg}_T^{(2)} \ge c\sqrt{T}\log T. \tag{13}$$

The hard family is already a threshold family. For $z \in [0,1]$ define

$$f_z(x) = \frac{x + \mathbf{1}\{z \ge x\}}{2}, \tag{14}$$

so the corresponding loss $1 - f_z$ belongs to $\mathcal{G}_{\mathrm{OSL}}$. Observing $f_z(x)$ is equivalent to observing the single threshold bit $\mathbf{1}\{z \ge x\}$.

Choose a hidden depth-L dyadic leaf center θ. For every dyadic ancestor $I_d(\theta)$, add a fixed-orientation Haar perturbation to the density of the threshold z. Integrating the Haar functions gives nonnegative tents $\tau_{I_d(\theta)}$ and the multiscale profile

$$H_\theta(x) = \sum_{d=0}^{L-1} \tau_{I_d(\theta)}(x). \tag{15}$$

Under the alternative indexed by θ, the mean reward is

$$\mu_\theta(x) = \frac{1}{2} + \frac{\delta}{2} H_\theta(x). \tag{16}$$

At the hidden point, every scale contributes coherently:

$$\mathbb{E}_\Theta H_\Theta(\Theta) = \frac{L}{4}. \tag{17}$$

The information available at a fixed query interval is much smaller. The key metric estimate is

$$\mathbb{E}_\Theta \big(H_\Theta(y) - H_\Theta(x)\big)^2 \le 64(y-x), \qquad 0 \le x < y \le 1. \tag{18}$$

Two simultaneous threshold queries $x < y$ reveal only which of the three bins $[0, x)$, $[x, y)$, and $[y, 1]$ contains the random threshold. Using (18), the KL divergence of this three-bin statistic from the uniform-threshold null is $O(\delta^2)$ on average over the hidden leaf, with no extra factor of L. Taking $L = \Theta(\log T)$ and $\delta = \Theta(T^{-1/2})$ keeps the total transcript information bounded while giving the hidden comparator a $\Theta(\delta L)$ per-round advantage. This yields (13). The full density construction, geometric lemmas, and adaptive KL argument appear in Section B.

Combining Theorems 3.1 and 4.1, the two-paid-point minimax regret is determined up to a factor $(\log T)^{1/4}$:

$$\Omega(\sqrt{T}\log T) \le R_T^{(2)}(\mathcal{G}_{\mathrm{OSL}}) \le O(\sqrt{T}(\log T)^{5/4}). \tag{19}$$

5 Full information

The multiscale logarithm disappears under full information because every dyadic comparison can be updated on every round.

Work with rewards $r_t = 1 - \ell_t$. Consider a dyadic node I of spatial width $w(I)$. Let $A_{t,I}$ and $B_{t,I}$ denote the current expected rewards of the left- and right-child subalgorithms. Their contrast

$$z_{t,I} = B_{t,I} - A_{t,I}$$

satisfies

$$-1 \le z_{t,I} \le w(I),$$

because moving right can improve reward by at most the node width, while rewards themselves remain in $[0, 1]$.

The local ingredient is an asymmetric two-action strategy.

Lemma 5.1 (Asymmetric two-action primitive)**.** *For every $\Delta \in (0,1]$, there is a full-information two-action strategy such that for every sequence $z_t \in [-1, \Delta]$,*

$$\left(\sum_{t=1}^{T} z_t\right)_+ - \sum_{t=1}^{T} q_t z_t \le \frac{1}{2}\sqrt{\Delta T}, \tag{20}$$

where q_t is the probability assigned to the action whose reward exceeds the baseline by z_t.

Place a dyadic grid of size at least T at the leaves of a complete binary tree and run an independent copy of Lemma 5.1 at every internal node with $\Delta = w(I)$. Sampling decisions down the tree produces the played leaf. After observing the full current loss function, every node receives its exact child-average contrast.

Theorem 5.1 (Full-information fixed-action OSL)**.** *There is a full-information learner satisfying*

$$\mathbb{E}\mathrm{Reg}_T \le \frac{\sqrt{T}}{2-\sqrt{2}} + 1. \tag{21}$$

Conversely, every randomized full-information learner has worst-case regret $\Omega(\sqrt{T})$. Hence

$$R_T^{\mathrm{full}}(\mathcal{G}_{\mathrm{OSL}}) = \Theta(\sqrt{T}). \tag{22}$$

For a fixed comparator leaf, only the nodes on its root-to-leaf path contribute. Their widths are $1, 1/2, 1/4, \ldots$, so the local costs sum geometrically:

$$\frac{1}{2}\sqrt{T}\sum_{d\ge 0} 2^{-d/2} = \frac{\sqrt{T}}{2-\sqrt{2}}.$$

The lower bound uses independent Rademacher choices between the two linear losses x and $1-x$. Proofs are in Section C.

6 Mechanism-design application: two-threshold pricing menus

Posted pricing is a canonical OSL problem. A buyer with value $v \in [0,1]$ generates revenue

$$r_v(p) = p\mathbf{1}\{p \le v\},$$

and the loss $\ell_v(p) = 1 - r_v(p)$ belongs to $\mathcal{G}_{\mathrm{OSL}}$.

A second abstract query is not automatically a legitimate pricing action: the seller cannot charge one price while freely probing another. The paid-average two-point protocol nevertheless has an exact incentive-compatible implementation.

Let $0 < q_1 < \cdots < q_m$ have weights $w_1, \ldots, w_m$ summing to one. Define cumulative allocation and payment levels

$$A_k = \sum_{j\le k} w_j, \qquad P_k = \sum_{j\le k} w_j q_j. \tag{23}$$

Offer the menu $(0,0), (A_1, P_1), \ldots, (A_m, P_m)$. The utility increment from tier $k-1$ to tier k is

$$w_k(v - q_k),$$

so a risk-neutral buyer chooses the largest tier whose threshold does not exceed v. The chosen tier therefore reveals all nested threshold bits in the support, and the payment is

$$P_k = \sum_{j=1}^{m} w_j q_j \mathbf{1}\{q_j \le v\} = \mathbb{E}_{Q\sim w}[r_v(Q)]. \tag{24}$$

For a simultaneous queried pair (B, C), use weights $1/2, 1/2$. The resulting menu has at most two positive thresholds, reveals the accept/reject bit at both prices, and earns exactly

$$\frac{r_v(B) + r_v(C)}{2}.$$

Thus it implements both the feedback and the paid objective of the abstract two-point model.

Corollary 6.1 (Two-threshold adversarial pricing)**.** *Against every oblivious buyer-value sequence, there is an incentive-compatible cumulative lottery menu with at most two positive price thresholds per round whose expected regret against the best fixed posted price is*

$$O\Big(\sqrt{T}\,(1+\log T)^{5/4}\Big). \tag{25}$$

The corollary follows by applying Theorem 3.1 to posted-price losses and replacing each queried pair by the corresponding cumulative menu. The menu identity is proved in Section D. Ordinary one-price adversarial posted pricing has minimax regret $\Theta(T^{2/3})$ (Kleinberg and Leighton, 2003); hence allowing a second paid threshold gives the same polynomial improvement as in the abstract OSL model.

The generic lower bound of Theorem 4.1 is a statement about arbitrary OSL losses. We do not claim here that it transfers to an unrestricted continuous two-threshold pricing learner.

7 Discussion

For adversarial one-sided Lipschitz learning against the best fixed action, the three feedback regimes now form a clear hierarchy:

$$\Theta(T^{2/3}) \quad\longrightarrow\quad \Big[\Omega(\sqrt{T}\log T),\ O(\sqrt{T}(\log T)^{5/4})\Big] \quad\longrightarrow\quad \Theta(\sqrt{T}). \tag{26}$$

A second paid observation is enough to change the horizon exponent from $2/3$ to $1/2$, but not enough to eliminate the multiscale logarithm. Full information can update all scales simultaneously and therefore pays only a geometric $O(\sqrt{T})$ sum.

The remaining gap is a factor $(\log T)^{1/4}$. In the upper-bound proof, feasibility and finite-step stability are already controlled by the hybrid barrier. The remaining loss enters through the static tree-energy estimate (11): the present proof gives $O(L^{3/2})$, while an $O(L)$ bound for the same or a comparable split schedule would yield the matching $O(\sqrt{T}\log T)$ regret. At the uniform distribution the corresponding energy is already $O(L)$, so the unresolved issue is the worst nonuniform geometry of the adaptive state.

The pricing implementation also clarifies why the paid-feedback model is not merely an oracle abstraction. The two observations can be generated by one incentive-compatible interaction with one buyer, with payment exactly matching the learner's average two-point objective. This provides a mechanism-design setting in which the value of the second observation is operational rather than informational only.

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A Proof of the two-point upper bound

This appendix proves Theorem 3.1. We first give the complete finite-grid learner, then establish the finite-chord potential inequality, global feasibility, the nonuniform tree-energy bound, and finally the regret calculation.

A.1 Tree notation and the explicit learner

Fix a grid with $K = 2^L$ leaves and let vL, vR denote the children of an internal node v. A node at depth d has spatial width $w_v = 2^{-d}$. For a leaf distribution p, let

$$p_v = \sum_{i \in v} p_i.$$

Set

$$\mu = 32\eta L, \qquad Z = \sum_{h=1}^{L} h^{-1/2}, \qquad \beta_d = \frac{1}{Z\sqrt{L-d}}. \tag{27}$$

Note that $\sum_{d=0}^{L-1} \beta_d = 1$ and $Z \le 2\sqrt{L}$.

Define

$$\Psi_0(p) = -\sum_{v} \sqrt{w_v p_v}, \qquad \Psi(p) = \Psi_0(p) - \mu \sum_{v \neq \text{root}} \log p_v, \tag{28}$$

and its conjugate on the simplex interior,

$$\Phi(s) = \max_{p \in \Delta_K^\circ} \{\langle s, p\rangle - \Psi(p)\}. \tag{29}$$

Strict convexity makes the optimizer unique; write $p = \nabla\Phi(s)$. Translation invariance on the simplex gives $\Phi(s + c\mathbf{1}) = \Phi(s) + c$.

For every internal node v, define

$$a_v = \frac{p_{vR}}{p_v}, \qquad \tau_v = -a_v \mathbf{1}_{vL} + (1 - a_v)\mathbf{1}_{vR}, \tag{30}$$

and

$$h_v(i) = \begin{cases} -p_i/p_{vL}, & i \in vL, \\ p_i/p_{vR}, & i \in vR, \\ 0, & i \notin v. \end{cases} \tag{31}$$

At depth d, let

$$q_v = \beta_d \min\{p_{vL}, p_{vR}\}, \qquad \kappa_v = \frac{\eta}{q_v}. \tag{32}$$

For the directional dual potential

$$\phi_v(u) = \Phi(s + u\tau_v),$$

define

$$c_v = \frac{\phi_v(\kappa_v w_v) - \phi_v(-\kappa_v)}{\kappa_v(1 + w_v)}, \tag{33}$$

$$B_v = \frac{w_v \phi_v(-\kappa_v) + \phi_v(\kappa_v w_v)}{1 + w_v} - \phi_v(0). \tag{34}$$

The actual paid marginal is

$$\pi = p + \sum_{v \text{ internal}} c_v h_v. \tag{35}$$

Finally, for a leaf i define

$$r_i = \pi_i - \frac{p_i}{2} \sum_{v \text{ ancestor of } i} \frac{q_v}{p_{\text{child}_v(i)}}, \tag{36}$$

where $\text{child}_v(i)$ is the child of v containing i.

On each round, independently of the current loss:

1. with probability q_v, choose node v, draw $A \sim p(\cdot \mid vL)$ and $B \sim p(\cdot \mid vR)$ independently, and randomize their order in the two query slots;
2. with probability r_i, query (i, i).

On a split at v, after observing

$$D = \ell_t(B) - \ell_t(A),$$

update

$$s_{t+1} = s_t - \kappa_v D \tau_v. \tag{37}$$

On a duplicate query, leave s unchanged. We initialize at $s_1 = 0$.

A.2 Finite-chord potential identity

We begin with the tree martingale decomposition.

Lemma A.1 (Tree martingale identity)**.** *For any vector $g \in \mathbb{R}^K$ and positive leaf distribution p, define*

$$d_v = \mathbb{E}_{p(\cdot|vR)} g - \mathbb{E}_{p(\cdot|vL)} g = ip h_v g.$$

Then

$$g - \langle p, g \rangle \mathbf{1} = \sum_{v \text{ internal}} d_v \tau_v. \tag{38}$$

Proof. For a leaf i, follow its root-to-leaf path. At an internal node v on this path, the contribution $d_v\tau_v(i)$ is the increment between the conditional mean at the child containing i and the conditional mean at v. Indeed, if $i \in vL$, then

$$\mathbb{E}[g \mid vL] - \mathbb{E}[g \mid v] = -\frac{p_{vR}}{p_v}\big(\mathbb{E}[g \mid vR] - \mathbb{E}[g \mid vL]\big),$$

and similarly for vR. Summing these conditional-mean increments telescopes from the root mean $\langle p, g\rangle$ to g_i. □

On a split at v, the OSL condition and the fact that every point in vL lies to the left of every point in vR imply

$$-w_v \le D \le 1. \tag{39}$$

Since $q_v\kappa_v = \eta$, Lemma A.1 gives

$$\mathbb{E}_t[s_{t+1} - s_t] = -\eta\big(\ell_t - \langle p, \ell_t\rangle \mathbf{1}\big). \tag{40}$$

The key point is to control the *finite* update $-\kappa_v D\tau_v$ rather than only its local quadratic approximation. By convexity, any $D \in [-w_v, 1]$ satisfies

$$\begin{aligned}\phi_v(-\kappa_v D) - \phi_v(0) &\le \frac{w_v + D}{1 + w_v}\phi_v(-\kappa_v) + \frac{1 - D}{1 + w_v}\phi_v(\kappa_v w_v) - \phi_v(0)\\ &= B_v - \kappa_v c_v D.\end{aligned} \tag{41}$$

Conditional on selecting v, $\mathbb{E}_t D = d_v$. Hence

$$\mathbb{E}_t[\Phi(s_{t+1}) - \Phi(s_t)] \le \sum_v q_v B_v - \eta\langle \pi - p, \ell_t\rangle, \tag{42}$$

because $\langle \pi - p, \ell_t\rangle = \sum_v c_v d_v$.

Lemma A.2 (Finite-grid regret identity)**.** *For every fixed interior comparator $u \in \Delta_K^\circ$,*

$$\mathbb{E}\sum_{t=1}^{T}\langle \pi_t - u, \ell_t\rangle \le \frac{\Psi(u) - \Psi(p_{\text{unif}})}{\eta} + \frac{1}{\eta}\sum_{t=1}^{T}\mathbb{E}\sum_v q_{t,v}B_{t,v}. \tag{43}$$

Proof. At every depth, Cauchy–Schwarz gives $\sum_v \sqrt{w_v p_v} \le 1$, with equality for the uniform leaf distribution. At each non-root depth, $-\sum_v \log p_v$ is likewise minimized by equal subtree masses. Thus p_{unif} minimizes Ψ and $\Phi(0) = -\Psi(p_{\text{unif}})$.

Summing (42), using Fenchel's inequality $\Phi(s_{T+1}) \ge \langle u, s_{T+1}\rangle - \Psi(u)$, and taking expectation in (40) gives

$$\eta\mathbb{E}\sum_t\langle p_t - u, \ell_t\rangle \le \Psi(u) - \Psi(p_{\text{unif}}) + \sum_t\mathbb{E}\sum_v q_{t,v}B_{t,v} - \eta\mathbb{E}\sum_t\langle \pi_t - p_t, \ell_t\rangle.$$

Moving the last term to the left proves the claim. □

A.3 The log barrier controls finite updates

Let $H = \nabla^2\Psi(p)$ and let G denote its inverse quadratic form on the simplex tangent space. Put

$$n_v = \tau_v^\top G \tau_v = \phi_v''(0), \qquad h = L - \mathrm{depth}(v). \tag{44}$$

Lemma A.3 (Subtree curvature from the barrier)**.** *For every internal node v,*

$$n_v \le \frac{a_v^2 p_{vL}^2 + (1-a_v)^2 p_{vR}^2}{\mu h} \le \frac{2\min\{p_{vL}, p_{vR}\}^2}{\mu h}. \tag{45}$$

Proof. Consider a subtree rooted at u with $j+1$ node levels including its root and leaves. The barrier terms alone imply that the minimum Hessian energy needed to create total displacement x in that subtree is at least

$$\frac{\mu(j+1)}{p_u^2} x^2. \tag{46}$$

This follows by induction. It is immediate at a leaf. For an internal node, the two child energies add; minimizing them under $x_L + x_R = x$ gives coefficient at least

$$\frac{\mu j}{p_{uL}^2 + p_{uR}^2} \ge \frac{\mu j}{p_u^2},$$

and the barrier at u adds μ/p_u^2.

Apply this to the two children of v, each with h levels, and drop all other nonnegative Hessian terms. The dual norm of the functional that assigns coefficient $-a_v$ to the left total displacement and $1-a_v$ to the right total displacement is at most the sum of the two reciprocal quadratic coefficients, giving the first inequality. Since $a_v = p_{vR}/p_v$ and $1 - a_v = p_{vL}/p_v$, both numerator terms equal $(p_{vL}p_{vR}/p_v)^2$, proving the second inequality. □

We next control curvature along the whole update chord.

Lemma A.4 (Finite-chord curvature)**.** *For any tangent vector z,*

$$|D^3\Psi(p)[z,z,z]| \le \frac{2}{\sqrt{\mu}} \left(z^\top H z\right)^{3/2}. \tag{47}$$

Consequently, along $\phi_v(u) = \Phi(s + u\tau_v)$,

$$\phi_v''(u) \le \frac{n_v}{\left(1 - |u|\sqrt{n_v/\mu}\right)^2} \quad \text{whenever } |u|\sqrt{n_v/\mu} < 1. \tag{48}$$

For the parameters in (27)–(32),

$$\phi_v''(u) \le 2n_v \qquad \text{for all } |u| \le \kappa_v. \tag{49}$$

Proof. For a square-root term $-\sqrt{w_u p_u}$, the magnitude of the third directional derivative is $(3/2)|z_u|/p_u$ times its Hessian contribution. For a barrier term $-\mu \log p_u$, the factor is $2|z_u|/p_u$. Since the barrier Hessian itself contains $\mu z_u^2/p_u^2$,

$$\frac{|z_u|}{p_u} \le \sqrt{\frac{z^\top H z}{\mu}}.$$

Summing the Hessian contributions proves (47).

By standard Legendre dual differentiation on the simplex tangent space, if z is the primal response to the direction τ_v, then

$$\phi_v'' = z^\top H z, \qquad \phi_v''' = -D^3\Psi[z,z,z].$$

Hence $|\phi_v'''| \le 2\phi_v''^{3/2}/\sqrt{\mu}$. Integrating the derivative of $1/\sqrt{\phi_v''}$ gives (48).

Let $m_v = \min\{p_{vL}, p_{vR}\}$. By Lemma A.3, for $|u| \le \kappa_v$,

$$|u|\sqrt{n_v/\mu} \le \frac{\eta}{\beta_d m_v}\frac{\sqrt{2}m_v}{\mu\sqrt{h}} = \frac{\sqrt{2}\eta Z}{\mu} \le \frac{\sqrt{2}}{16\sqrt{L}} < \frac{1}{4}.$$

Substitution into (48) gives (49). □

Because $\phi_v'(0) = \langle p, \tau_v\rangle = 0$, (49) implies the endpoint quadratic bounds

$$0 \le \phi_v(x) - \phi_v(0) \le n_v x^2 \qquad (|x| \le \kappa_v).$$

Using (33)–(34) therefore gives

$$|c_v| \le \kappa_v n_v, \qquad B_v \le \kappa_v^2 w_v n_v. \tag{50}$$

Lemma A.5 (Global feasibility)**.** *The sampling rule is valid at every positive state. In particular,*

$$r_i \ge \frac{p_i}{4} > 0 \qquad \textit{for every leaf } i, \tag{51}$$

and both query slots have marginal π.

Proof. Along the root-to-leaf path of i, the split events consume from either query slot at most

$$\sum_{v \prec i} \frac{q_v}{2p_{\mathrm{child}_v(i)}} \le \frac{1}{2}\sum_{d=0}^{L-1}\beta_d = \frac{1}{2}. \tag{52}$$

On the same path, Lemma A.3 and (50) give

$$\sum_{v \prec i} \frac{|c_v|}{p_{\mathrm{child}_v(i)}} \le \frac{2\eta}{\mu}\sum_{d=0}^{L-1}\frac{1}{\beta_d(L-d)} = \frac{2\eta Z^2}{\mu} \le \frac{1}{4}. \tag{53}$$

Here $Z^2 \le 4L$ and $\mu = 32\eta L$.

From (35) and (31), the correction to coordinate i divided by p_i has magnitude at most (53). Combining with (52) in (36) proves (51).

Every h_v has zero coordinate sum, so $\sum_i \pi_i = 1$. For a split at v, randomizing endpoint order contributes $q_v p_i/(2p_{\mathrm{child}_v(i)})$ to either slot at leaf i; the duplicate event contributes r_i. The definition of r_i makes their sum exactly π_i. Summing over i also gives $\sum_i r_i = 1 - \sum_v q_v$, so the event probabilities sum to one. □

A.4 A nonuniform square-root tree energy inequality

We now use only Ψ_0. Let G_0 denote its inverse-Hessian quadratic form on the simplex tangent space and define

$$\mathcal{A}_0(p) = \sum_{v \text{ internal}} \frac{w_v \tau_v^\top G_0 \tau_v}{\beta_{\mathrm{depth}(v)} \min\{p_{vL}, p_{vR}\}}. \tag{54}$$

Theorem A.1 (Nonuniform tree energy)**.** *For every strictly positive leaf distribution p,*

$$\mathcal{A}_0(p) \le 128 L^{3/2}. \tag{55}$$

Proof. For each node let

$$k_v = \frac{\sqrt{w_v}}{4 p_v^{3/2}}. \tag{56}$$

This is the coefficient of the square-root Hessian term acting on the total displacement of subtree v. Let R_v be the minimum Hessian energy inside the subtree for unit total displacement at its root. Direct minimization gives the effective-resistance recursion

$$R_i = k_i \quad \text{at leaves}, \qquad R_v = k_v + (R_{vL}^{-1} + R_{vR}^{-1})^{-1}. \tag{57}$$

Define

$$\rho_v = \sqrt{w_v p_v}, \qquad \gamma_v = \frac{k_v}{R_v} \in (0, 1], \qquad t_v = \rho_v \gamma_v. \tag{58}$$

At leaves $\gamma_i = 1$. At any fixed depth, Cauchy–Schwarz gives

$$\sum_{v:\, \mathrm{depth}(v)=d} \rho_v \le 1. \tag{59}$$

Fix an internal v and write $a = p_{vL}/p_v$, $b = p_{vR}/p_v$, so $a + b = 1$. From (57),

$$\gamma_v = \frac{\nu}{1+\nu}, \qquad \nu = \sqrt{2}\big(a^{3/2}\gamma_{vL} + b^{3/2}\gamma_{vR}\big). \tag{60}$$

Let

$$s = \frac{t_{vL} + t_{vR}}{\rho_v}, \qquad \delta = |a - b|.$$

A direct substitution gives

$$\nu - s = (a - b)\frac{t_{vL} - t_{vR}}{\rho_v}, \qquad \nu \le \sqrt{2}. \tag{61}$$

If $\nu \ge s$, then $\nu - s \le \delta s \le \delta\nu$; if $\nu < s$, the same upper bound is trivial. Put $\kappa_0 = (1 + \sqrt{2})^{-1}$. Since $\nu^2/(1+\nu) \ge \kappa_0 \nu^2$,

$$\begin{aligned} \frac{\nu}{1+\nu} - s &\le \delta\nu - \kappa_0 \nu^2 \\ &\le \frac{\delta^2}{2\kappa_0} - \frac{\kappa_0}{2}\nu^2 \le \frac{\delta^2}{2\kappa_0} - \frac{\kappa_0}{2}\gamma_v^2. \end{aligned} \tag{62}$$

Moreover,

$$\delta^2 \le 8\left(1 - \frac{\sqrt{a} + \sqrt{b}}{\sqrt{2}}\right) = 8\frac{\rho_v - \rho_{vL} - \rho_{vR}}{\rho_v}. \tag{63}$$

For example, the first inequality follows from $1-4ab=(1-2\sqrt{ab})(1+2\sqrt{ab})$ and $1-r\ge(1-r^2)/2$ with $r=(\sqrt{a}+\sqrt{b})/\sqrt{2}$.

Multiplying (62) by ρ_v and using (63) yields

$$t_v+c_0\frac{t_v^2}{\rho_v}\le t_{vL}+t_{vR}+C_0(\rho_v-\rho_{vL}-\rho_{vR}), \tag{64}$$

where

$$c_0=\frac{1}{2(1+\sqrt{2})},\qquad C_0=4(1+\sqrt{2}).$$

Summing (64) over all internal nodes telescopes both the t and ρ terms. Since $\rho_{\text{root}}=1$, $t_i=\rho_i$ at leaves, and $\sum_i\rho_i\le 1$,

$$\sum_{v\ \text{internal}}\frac{t_v^2}{\rho_v}\le\frac{C_0}{c_0}.$$

Adding the leaves gives the convenient bound

$$\sum_{v\neq\text{root}}\frac{t_v^2}{\rho_v}\le 1+8(1+\sqrt{2})^2<48. \tag{65}$$

It remains to connect this effective subtree energy to the contrast norms in (54). Let $C_u=1/R_u$. Dropping all Hessian terms outside the two child subtrees and using the dual variational characterization of G_0 gives

$$\tau_v^\top G_0\tau_v\le\left(\frac{p_{vR}}{p_v}\right)^2C_{vL}+\left(\frac{p_{vL}}{p_v}\right)^2C_{vR}. \tag{66}$$

From (56)–(58),

$$t_u=\frac{w_uC_u}{4p_u}. \tag{67}$$

If, for example, $p_{vL}\le p_{vR}$, substituting (67) into (66) gives

$$\frac{w_v\tau_v^\top G_0\tau_v}{p_{vL}}\le 8\left[\left(\frac{p_{vR}}{p_v}\right)^2t_{vL}+\frac{p_{vL}p_{vR}}{p_v^2}t_{vR}\right]\le 8(t_{vL}+t_{vR}). \tag{68}$$

The other case is symmetric. Therefore

$$\begin{aligned}\mathcal{A}_0(p)&\le 8\sum_{u\neq\text{root}}\frac{t_u}{\beta_{\text{depth}(u)-1}}\\&\le 8\sqrt{48}\left(\sum_{u\neq\text{root}}\frac{\rho_u}{\beta^2_{\text{depth}(u)-1}}\right)^{1/2} &(69)\\&\le 8\sqrt{48}\,Z\sqrt{\frac{L(L+1)}{2}}\le 128L^{3/2}, &(70)\end{aligned}$$

where (69) uses (65), and the next line uses (59). This proves (55). □

A.5 Completion of the regret bound

Adding the convex barrier only increases the Hessian, so its inverse quadratic form satisfies $G \preceq G_0$ on the tangent space. By (50), (32), and Theorem A.1,

$$\sum_v q_v B_v \le \eta^2 \sum_v \frac{w_v \tau_v^\top G \tau_v}{q_v} \le 128\eta^2 L^{3/2}. \tag{71}$$

Fix a comparator leaf i and smooth it to

$$u = (1-\varepsilon)e_i + \varepsilon p_{\mathrm{unif}}.$$

Every subtree mass satisfies $u_v \ge \varepsilon w_v$. The square-root part of the comparator divergence is at most $L+1$. For the barrier part,

$$\mu \sum_{v \ne \mathrm{root}} \log \frac{w_v}{u_v} \le 2\mu K \log(1/\varepsilon),$$

because a complete binary tree has fewer than $2K$ non-root nodes. Hence

$$\Psi(u) - \Psi(p_{\mathrm{unif}}) \le L + 1 + 2\mu K \log(1/\varepsilon). \tag{72}$$

The smoothing itself costs at most εT against the pure leaf comparator. Substituting (71) and (72) into Lemma A.2, with $\mu = 32\eta L$, gives

$$\mathbb{E}\mathrm{Reg}_T^{\mathrm{grid}} \le \frac{L+1}{\eta} + 64KL\log(1/\varepsilon) + 128\eta T L^{3/2} + \varepsilon T. \tag{73}$$

To pass to the continuous comparator, round it downward to the grid, paying at most T/K by (4). Set $\varepsilon = 1/T$ and let K be the smallest power of two at least

$$\max\left\{2, \frac{\sqrt{T}}{1+\log T}\right\}.$$

Then $L = \Theta(1+\log T)$. Choose

$$\eta = \frac{1}{\sqrt{T}\,(L+1)^{1/4}}.$$

The first and third terms of (73) are $O(\sqrt{T}(1+\log T)^{5/4})$, while the barrier term and T/K are $O(\sqrt{T}(1+\log T))$, and $\varepsilon T = 1$. Enlarging the universal constant handles small T. This proves Theorem 3.1.

B Proof of the two-point lower bound

This appendix proves Theorem 4.1. The proof works with a threshold reward family and a hidden dyadic path.

B.1 Threshold family and hidden alternatives

For $z \in [0,1]$ define

$$f_z(x) = \frac{x + \mathbf{1}\{z \geq x\}}{2}, \qquad x \in [0,1]. \tag{74}$$

For $x \leq y$,

$$f_z(y) - f_z(x) \leq \frac{y-x}{2} \leq y - x,$$

so the loss $\ell_z = 1 - f_z$ belongs to $\mathcal{G}_{\mathrm{OSL}}$. Observing either $f_z(x)$ or $\ell_z(x)$ is equivalent to observing the threshold bit $\mathbf{1}\{z \geq x\}$.

Fix

$$L = \left\lfloor \frac{1}{2} \log_2 T \right\rfloor, \qquad K = 2^L, \qquad \delta = \frac{c_0}{\sqrt{T}}, \qquad c_0 = \frac{1}{128\sqrt{2}}. \tag{75}$$

Let

$$\Theta_L = \left\{ \frac{j + 1/2}{K} : j = 0, \ldots, K-1 \right\}$$

be the centers of the depth-L dyadic cells. For $\theta \in \Theta_L$, let $I_d(\theta)$ be its unique dyadic ancestor of depth d, with width $w_d = 2^{-d}$.

For a dyadic interval $I = [a, a+w]$ with left and right halves I_L, I_R, define the fixed-orientation Haar function

$$\psi_I(z) = \frac{1}{w}\big(\mathbf{1}\{z \in I_L\} - \mathbf{1}\{z \in I_R\}\big) \tag{76}$$

and its cumulative integral

$$\tau_I(x) = \int_0^x \psi_I(z)\, dz = \mathbf{1}\{x \in I\} \frac{\min\{x - a, a + w - x\}}{w}. \tag{77}$$

Thus $0 \leq \tau_I \leq 1/2$, τ_I vanishes outside I, and it is $1/w$-Lipschitz.

Define

$$G_\theta(z) = \sum_{d=0}^{L-1} \psi_{I_d(\theta)}(z), \qquad H_\theta(x) = \sum_{d=0}^{L-1} \tau_{I_d(\theta)}(x). \tag{78}$$

The alternative indexed by θ draws Z_t independently from density

$$\rho_\theta(z) = 1 - \delta G_\theta(z). \tag{79}$$

Every Haar term integrates to zero. Also

$$|G_\theta(z)| \leq \sum_{d=0}^{L-1} 2^d = K - 1.$$

Since $K \leq \sqrt{T}$ and $\delta K \leq c_0 < 1/4$,

$$\frac{3}{4} \leq \rho_\theta(z) \leq \frac{5}{4},$$

so (79) is a valid density. Let $\rho_0 \equiv 1$ be the null. Its CDF under alternative θ is

$$F_\theta(x) = x - \delta H_\theta(x),$$

and therefore

$$\mu_\theta(x) := \mathbb{E}_{Z \sim \rho_\theta} f_Z(x) = \frac{1}{2} + \frac{\delta}{2} H_\theta(x). \tag{80}$$

B.2 Geometry of the hidden tents

Lemma B.1 (Dyadic tent geometry)**.** *If Θ is uniform on Θ_L, then:*

1. $0 \le H_\theta(x) \le L/2$ *for every* θ, x*;*
2. *for every fixed* $x \in [0,1]$, $\mathbb{E}_\Theta H_\Theta(x) < 1$*;*
3. $\mathbb{E}_\Theta H_\Theta(\Theta) = L/4$.

Proof. The first claim follows from $0 \le \tau_I \le 1/2$. At depth d, $I_d(\Theta)$ is uniform among the 2^d intervals of width w_d. Only the interval containing a fixed x can contribute, so

$$\mathbb{E}_\Theta \tau_{I_d(\Theta)}(x) \le \frac{w_d}{2}.$$

Summing gives

$$\mathbb{E}_\Theta H_\Theta(x) \le \frac{1}{2} \sum_{d=0}^{L-1} 2^{-d} < 1.$$

For the third claim, condition on a depth-d interval. The depth-L leaf centers are evenly spaced inside it. Their average tent height is exactly $1/4$ because, for even $n = 2^{L-d}$,

$$\frac{1}{n} \sum_{j=0}^{n-1} \min\left\{ \frac{j+1/2}{n}, 1 - \frac{j+1/2}{n} \right\} = \frac{1}{4}.$$

Every depth therefore contributes $1/4$ on average. □

Lemma B.2 (Square-increment bound)**.** *For every $0 \le x < y \le 1$,*

$$\mathbb{E}_\Theta \big(H_\Theta(y) - H_\Theta(x) \big)^2 \le 64(y - x). \tag{81}$$

In particular,

$$\mathbb{E}_\Theta H_\Theta(x)^2 \le 64x, \qquad \mathbb{E}_\Theta H_\Theta(x)^2 \le 64(1-x). \tag{82}$$

Proof. Put $r = y - x$. At depth d, with width $w = 2^{-d}$, let

$$X_d(\Theta) = \tau_{I_d(\Theta)}(y) - \tau_{I_d(\Theta)}(x).$$

At most the depth-d intervals containing x or y can contribute, so X_d is nonzero with probability at most $2w$. Since a width-w tent is $1/w$-Lipschitz and bounded by $1/2$,

$$|X_d(\Theta)| \le \min\left\{ \frac{r}{w}, \frac{1}{2} \right\}.$$

Hence

$$\|X_d\|_{L_2(\Theta)} \le \sqrt{2w} \min\left\{ \frac{r}{w}, \frac{1}{2} \right\}. \tag{83}$$

By Minkowski,

$$\|H_\Theta(y) - H_\Theta(x)\|_2 \le \sum_d \|X_d\|_2.$$

Split the dyadic widths into $w \ge r$ and $w < r$. For $w \ge r$, (83) is at most $\sqrt{2}\, r/\sqrt{w}$; for $w < r$ it is at most $\sqrt{w/2}$. Both dyadic sums are geometric and together are at most $8\sqrt{r}$. Squaring proves (81). Every tent vanishes at 0 and 1, so setting $(x, y) = (0, x)$ and $(x, 1)$ gives (82). □

B.3 Information in two simultaneous queries

Fix two queries $0 \le x < y \le 1$. The two threshold bits reveal exactly which of the three bins

$$[0, x), \qquad [x, y), \qquad [y, 1]$$

contains Z. Under the null the bin probabilities are

$$q_0 = (x, y - x, 1 - y).$$

Under alternative θ they are

$$q_\theta = \big(x - \delta H_\theta(x), y - x + \delta H_\theta(x) - \delta H_\theta(y), 1 - y + \delta H_\theta(y)\big).$$

Because $\rho_\theta \ge 3/4$ pointwise, every nonempty alternative bin has probability at least $3/4$ of its null probability. Therefore

$$\begin{aligned} D_{\mathrm{KL}}(q_0 \| q_\theta) &\le \chi^2(q_0 \| q_\theta) \\ &\le \frac{4\delta^2}{3} \left(\frac{H_\theta(x)^2}{x} + \frac{(H_\theta(y) - H_\theta(x))^2}{y - x} + \frac{H_\theta(y)^2}{1 - y} \right), \end{aligned} \tag{84}$$

with zero-denominator terms omitted. Averaging over Θ and applying Lemma B.2 bounds each normalized term by 64, hence

$$\frac{1}{K} \sum_{\theta \in \Theta_L} D_{\mathrm{KL}}(q_0 \| q_\theta) \le 256\delta^2. \tag{85}$$

B.4 Adaptive lower-bound argument

Proof of Theorem 4.1. Fix an arbitrary randomized learner. Let P_0 be the law of its complete interaction transcript under independent uniform thresholds Z_t, and P_θ the corresponding law when the thresholds are independent with density ρ_θ.

Let (B_t, C_t) be the simultaneous queried pair. For analysis only, draw A_t uniformly from the two endpoints conditional on (B_t, C_t). Then

$$\mathbb{E}[f_{Z_t}(A_t) \mid B_t, C_t, Z_t] = \frac{1}{2}\big(f_{Z_t}(B_t) + f_{Z_t}(C_t)\big),$$

so A_t has exactly the learner's expected paid reward. Adding A_t to the transcript contributes no relative entropy because it is generated by the same conditional kernel under every environment.

Conditional on any history under P_0, the next query kernel is identical under all alternatives and is chosen before the current threshold is observed. The adaptive chain rule and (85) imply

$$\frac{1}{K} \sum_{\theta \in \Theta_L} D_{\mathrm{KL}}(P_0 \| P_\theta) \le 256 T \delta^2. \tag{86}$$

Pinsker and Jensen give

$$\frac{1}{K} \sum_{\theta} d_{\mathrm{TV}}(P_0, P_\theta) \le \sqrt{\frac{1}{2K} \sum_{\theta} D_{\mathrm{KL}}(P_0 \| P_\theta)} \le \sqrt{128 T \delta^2} = \frac{1}{16}. \tag{87}$$

For each θ, define

$$S_\theta = \sum_{t=1}^{T} H_\theta(A_t).$$

By Lemma B.1, $0 \le S_\theta \le TL/2$. Under the null, the learner's transcript is independent of a uniformly sampled Θ, so

$$\frac{1}{K}\sum_{\theta} \mathbb{E}_0 S_\theta \le T. \tag{88}$$

Total-variation comparison and (87) yield

$$\begin{aligned}\frac{1}{K}\sum_{\theta} \mathbb{E}_\theta S_\theta &\le \frac{1}{K}\sum_{\theta} \mathbb{E}_0 S_\theta + \frac{TL}{2K}\sum_{\theta} d_{\mathrm{TV}}(P_0, P_\theta) \\ &\le T + \frac{TL}{32}.\end{aligned} \tag{89}$$

Under alternative θ, compare with the fixed action $x_\theta = \theta$. By (80),

$$\mathbb{E}_\theta \sum_{t=1}^{T} \big(f_{Z_t}(x_\theta) - f_{Z_t}(A_t)\big) = \frac{\delta}{2}\big(TH_\theta(x_\theta) - \mathbb{E}_\theta S_\theta\big).$$

Averaging over θ and using Lemma B.1 together with (89),

$$\begin{aligned}\frac{1}{K}\sum_{\theta} \mathbb{E}_\theta \mathrm{Reg}_T^{(2)} &\ge \frac{\delta T}{2}\left(\frac{L}{4} - 1 - \frac{L}{32}\right) \\ &\ge \frac{\delta TL}{16}\end{aligned}$$

for $L \ge 11$. Since $\delta = c_0/\sqrt{T}$ and $L = \Theta(\log T)$, the lower bound is $\Omega(\sqrt{T}\log T)$.

The random choice of θ and the entire threshold sequence may be made before interaction, so this is an oblivious randomized adversary. Averaging implies that some deterministic oblivious sequence in its support attains the same-order expected regret against the fixed randomized learner. □

C Proofs for full information

C.1 Asymmetric two-action primitive

Proof of Lemma 5.1. Let Z equal Δ with probability $1/(1+\Delta)$ and -1 with probability $\Delta/(1+\Delta)$. Then $\mathbb{E}Z = 0$ and $\mathbb{E}Z^2 = \Delta$. Define

$$\Phi_t(s) = \mathbb{E}\left[\left(s + \sum_{\tau=t+1}^{T} Z_\tau\right)_+\right],$$

where the remaining perturbations are independent copies of Z. If $S_{t-1} = \sum_{s<t} z_s$, put

$$q_t = \frac{\Phi_t(S_{t-1}+\Delta) - \Phi_t(S_{t-1}-1)}{1+\Delta}. \tag{90}$$

Monotonicity and one-Lipschitzness of Φ_t imply $q_t \in [0,1]$. For any $z \in [-1, \Delta]$, convexity gives

$$\Phi_t(S+z) \le \frac{z+1}{1+\Delta}\Phi_t(S+\Delta) + \frac{\Delta - z}{1+\Delta}\Phi_t(S-1).$$

Comparing with

$$\Phi_{t-1}(S) = \frac{1}{1+\Delta}\Phi_t(S+\Delta) + \frac{\Delta}{1+\Delta}\Phi_t(S-1)$$

yields

$$\Phi_t(S+z) - q_t z \le \Phi_{t-1}(S).$$

Telescoping gives

$$\left(\sum_t z_t\right)_+ - \sum_t q_t z_t \le \Phi_0(0).$$

Since the perturbation sum is centered,

$$\Phi_0(0) = \frac{1}{2}\mathbb{E}\left|\sum_{t=1}^{T} Z_t\right| \le \frac{1}{2}\sqrt{T\mathbb{E}Z^2} = \frac{1}{2}\sqrt{\Delta T}.$$

□

C.2 Dyadic full-information learner

Proof of Theorem 5.1, upper bound. Let K be the smallest power of two at least T and place the grid points $0, 1/K, \dots, (K-1)/K$ at the leaves of a complete binary tree. For an internal node I, let $w(I) = 2^{-\mathrm{depth}(I)}$ be its spatial width. Run Lemma 5.1 at node I with $\Delta = w(I)$. Sampling the node decisions along a root-to-leaf path gives the played leaf.

After the full grid reward vector is observed, let $A_{t,I}$ and $B_{t,I}$ be the current expected rewards of the left- and right-child subalgorithms before update, and feed

$$z_{t,I} = B_{t,I} - A_{t,I}.$$

For every left leaf x and right leaf y inside I, (5) gives

$$r_t(y) - r_t(x) \le y - x \le w(I),$$

while rewards lie in $[0,1]$. Hence $z_{t,I} \in [-1, w(I)]$, as required by the local primitive.

Fix a comparator leaf and follow its root-to-leaf path. The local regrets telescope through the hierarchy. At depth d, the path node has width 2^{-d}, so its local regret is at most

$$\frac{1}{2}\sqrt{2^{-d}T}.$$

Summing the geometric series gives

$$\frac{1}{2}\sqrt{T}\sum_{d\ge 0} 2^{-d/2} = \frac{\sqrt{T}}{2-\sqrt{2}}.$$

Finally, rounding an arbitrary continuous comparator downward to the grid costs at most $T/K \le 1$ by (4). □

Proof of Theorem 5.1, lower bound. Draw independent Rademacher variables $\epsilon_1, \dots, \epsilon_T$ and define

$$\ell_t(x) = \frac{1}{2} - \epsilon_t \left(x - \frac{1}{2} \right). \tag{91}$$

When $\epsilon_t = 1$ this is $1 - x$, and when $\epsilon_t = -1$ it is x; both belong to $\mathcal{G}_{\mathrm{OSL}}$. Since ϵ_t is independent of the learner's action conditional on the past,

$$\mathbb{E}[\ell_t(X_t) \mid \mathcal{H}_{t-1}] = \frac{1}{2}.$$

Writing $S_T = \sum_t \epsilon_t$, the best fixed endpoint has cumulative loss

$$\frac{T}{2} - \frac{|S_T|}{2}.$$

Thus the expected regret is $\mathbb{E}|S_T|/2 = \Omega(\sqrt{T})$. Yao's principle yields a deterministic oblivious sequence for every randomized learner. □

D Proof of the pricing menu identity

Let $0 < q_1 < \cdots < q_m$ and let weights $w_j \geq 0$ sum to one. Define

$$A_k = \sum_{j \leq k} w_j, \qquad P_k = \sum_{j \leq k} w_j q_j,$$

and offer the menu

$$(0, 0), (A_1, P_1), \dots, (A_m, P_m).$$

A buyer of value v has utility

$$u_k(v) = vA_k - P_k.$$

The utility difference between consecutive tiers is

$$u_k(v) - u_{k-1}(v) = w_k(v - q_k).$$

With ties broken toward the larger allocation, the buyer therefore chooses the largest k with $q_k \leq v$. The chosen tier reveals all nested threshold bits $\mathbf{1}\{q_j \leq v\}$. Its payment is

$$P_k = \sum_{j=1}^{m} w_j q_j \mathbf{1}\{q_j \leq v\} = \mathbb{E}_{Q \sim w}[Q\mathbf{1}\{Q \leq v\}],$$

which is exactly the expected posted-price revenue under threshold distribution w.

For two queried prices B, C, choose weights $1/2, 1/2$, combining them if $B = C$. The menu has at most two positive thresholds, reveals the response at both prices, and earns

$$\frac{B\mathbf{1}\{B \leq v\} + C\mathbf{1}\{C \leq v\}}{2}.$$

Hence replacing each simultaneous queried pair of the OSL learner by this menu preserves both its feedback and its paid objective. Applying Theorem 3.1 to the posted-price losses proves Corollary 6.1.